

- Generally, do not use with MAOIs, including 14 days after MAOIs are stopped; do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing protriptyline
- Use with caution in patients with history of seizures, urinary retention, angle-closure glaucoma, hyperthyroidism
- TCAs can increase QTc interval, especially at toxic doses, which can be attained not only by overdose but also by combining with drugs that inhibit TCA metabolism via CYP450 2D6, potentially causing torsade de pointes-type arrhythmia or sudden death
- Because TCAs can prolong QTc interval, use with caution in patients who have bradycardia or who are taking drugs that can induce bradycardia (e.g., beta blockers, calcium channel blockers, clonidine, digitalis)
- Because TCAs can prolong QTc interval, use with caution in patients who have hypokalemia and/or hypomagnesemia or who are taking drugs that can induce hypokalemia and/or magnesemia (e.g., diuretics, stimulant laxatives, intravenous amphotericin B, glucocorticoids, tetracosactide)
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies

- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is recovering from myocardial infarction
- If patient is taking agents capable of significantly prolonging QTc interval (e.g., pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin)
- If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- If patient is taking drugs that inhibit TCA metabolism, including CYP450 2D6 inhibitors, except by an expert
- If there is reduced CYP450 2D6 function, such as patients who are poor 2D6 metabolizers, except by an expert and at low doses
- If there is a proven allergy to protriptyline

Special Populations

Renal Impairment

- Use with caution; may need to lower dose

- Patient may need to be monitored closely

Hepatic Impairment

- Use with caution; may need to lower dose
- Patient may need to be monitored closely

Cardiac Impairment

- Baseline ECG is recommended
- TCAs have been reported to cause arrhythmias, prolongation of conduction time, orthostatic hypotension, sinus tachycardia, and heart failure, especially in the diseased heart
- Myocardial infarction and stroke have been reported with TCAs
- TCAs produce QTc prolongation, which may be enhanced by the existence of bradycardia, hypokalemia, congenital or acquired long QTc interval, which should be evaluated prior to administering protriptyline
- Use with caution if treating concomitantly with a medication likely to produce prolonged bradycardia, hypokalemia, slowing of intracardiac conduction, or prolongation of the QTc interval
- Avoid TCAs in patients with a known history of QTc prolongation, recent acute myocardial infarction, and uncompensated heart failure
- TCAs may cause a sustained increase in heart rate in patients with ischemic heart disease and may worsen (decrease) heart rate variability, an independent risk of mortality in cardiac populations

- Since SSRIs may improve (increase) heart rate variability in patients following a myocardial infarct and may improve survival as well as mood in patients with acute angina or following a myocardial infarction, these are more appropriate agents for cardiac population than tricyclic/tetracyclic antidepressants

* Risk/benefit ratio may not justify use of TCAs in cardiac impairment

Elderly

- Baseline ECG is recommended for patients over age 50
- May be more sensitive to anticholinergic, cardiovascular, hypotensive, and sedative effects
- Recommended dose is between 15–20 mg/day; doses >20 mg/day require close monitoring of patient
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment

- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Not recommended for use under age 12
- Not intended for use under age 6
- Several studies show lack of efficacy of TCAs for depression
- Some cases of sudden death have occurred in children taking TCAs
- Recommended dose: 15–20 mg/day

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Crosses the placenta
- Adverse effects have been reported in infants whose mothers took a TCA (lethargy, withdrawal symptoms, fetal malformations)
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no

treatment (recurrence of depression, maternal health, infant bonding) to the mother and child

- For many patients this may mean continuing treatment during pregnancy
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Severe or treatment-resistant depression
- Withdrawn, anergic patients

Potential Disadvantages

- Pediatric, geriatric, and cardiac patients
- Patients concerned with weight gain
- Patients noncompliant with 3–4 times daily dosing

Primary Target Symptoms

- Depressed mood

Pearls

- TCAs are no longer generally considered a first-line treatment option for depression because of their side effect profile
- TCAs continue to be useful for severe or treatment-resistant depression
- * Has some potential advantages for withdrawn, anergic patients
- * May have a more rapid onset of action than some other TCAs
- * May aggravate agitation and anxiety more than some other TCAs
- * May have more anticholinergic side effects, hypotension, and tachycardia than some other TCAs
- Noradrenergic reuptake inhibitors such as protriptyline can be used as a second-line treatment for smoking cessation, cocaine

dependence, and attention deficit disorder

- TCAs may aggravate psychotic symptoms
- Alcohol should be avoided because of additive CNS effects
- Underweight patients may be more susceptible to adverse cardiovascular effects
- Children, patients with inadequate hydration, and patients with cardiac disease may be more susceptible to TCA-induced cardiotoxicity than healthy adults
- For the expert only: a heroic treatment (but potentially dangerous) for severely treatment-resistant patients is to give simultaneously with MAOIs for patients who fail to respond to numerous other antidepressants, but generally recommend a different TCA than protriptyline for this use
- If this option is elected, start the MAOI with the tricyclic/tetracyclic antidepressant simultaneously at low doses after appropriate drug washout, then alternately increase doses of these agents every few days to a week as tolerated
- Although very strict dietary and concomitant drug restrictions must be observed to prevent hypertensive crises and serotonin syndrome, the most common side effects of MAOI and tricyclic/tetracyclic antidepressant combinations may be weight gain and orthostatic hypotension
- Patients on TCAs should be aware that they may experience symptoms such as photosensitivity or blue-green urine

- SSRIs may be more effective than TCAs in women, and TCAs may be more effective than SSRIs in men
- Since tricyclic/tetracyclic antidepressants are substrates for CYP450 2D6, and 7% of the population (especially Caucasians) may have a genetic variant leading to reduced activity of 2D6, such patients may not safely tolerate normal doses of tricyclic/tetracyclic antidepressants and may require dose reduction
- Phenotypic testing may be necessary to detect this genetic variant prior to dosing with a tricyclic/tetracyclic antidepressant, especially in vulnerable populations such as children, elderly, cardiac populations, and those on concomitant medications
- Patients who seem to have extraordinarily severe side effects at normal or low doses may have this phenotypic CYP450 2D6 variant and require low doses or switching to another antidepressant not metabolized by 2D6

Suggested Reading

Anderson IM . Selective serotonin reuptake inhibitors versus tricyclic antidepressants: a meta-analysis of efficacy and tolerability . *J Aff Disorders* 2000 ;**58** :19 –36 .

Anderson IM . Meta-analytical studies on new antidepressants . *Br Med Bull* 2001 ;**57** :161–78 .

.....

Rudorfer MV , Potter WZ . Metabolism of tricyclic antidepressants . *Cell Mol Neurobiol* 1999 ;**19** (3):373 –409 .

Quazepam

Therapeutics

Brands

- Doral
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (hypnotic)

Commonly Prescribed for

(bold for FDA approved)

- **Short-term treatment of insomnia**
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness

- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth
- Rebound insomnia when withdrawing from long-term treatment



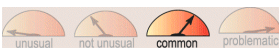
Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain

- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, only take quazepam if planning to have a full night's sleep

- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Switch to a non-benzodiazepine hypnotic
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 15 mg/day at bedtime

Dosage Forms

- Tablet 15 mg

How to Dose

- 15 mg/day at bedtime; increase to 30 mg/day if ineffective; maximum dose 30 mg/day



Dosing Tips

- Use lowest possible effective dose and assess need for continued treatment regularly

- Quazepam should generally not be prescribed in quantities greater than a 1-month supply
- Patients with lower body weights may require lower doses
- Risk of dependence may increase with dose and duration of treatment

Overdose

- No death reported in monotherapy; sedation, respiratory depression, poor coordination, confusion, coma

Long-Term Use

- Not generally intended for use beyond 4 weeks
- * Because of its relatively longer half-life, quazepam may cause some daytime sedation and/or impaired motor/cognitive function, and may do so progressively over time

Habit Forming

- Quazepam is a Schedule IV drug
- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- If taken for more than a few weeks, taper to reduce chances of withdrawal effects
- Patients with seizure history may seize upon sudden withdrawal
- Rebound insomnia may occur the first 1–2 nights after stopping
- For patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

Pharmacokinetics

- Half-life 25–41 hours
- Active metabolite
- Metabolized in part by CYP450 3A4



Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)
- Effects of quazepam may be increased by CYP450 3A4 inhibitors such as nefazodone or fluvoxamine



Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea
- Quazepam should be administered only at bedtime

Do Not Use

- If patient is pregnant

- If patient has angle-closure glaucoma
- If there is a proven allergy to quazepam or any benzodiazepine

Special Populations

Renal Impairment

- Recommended dose: 7.5 mg/day

Hepatic Impairment

- Recommended dose: 7.5 mg/day

Cardiac Impairment

- Benzodiazepines have been used to treat insomnia associated with acute myocardial infarction

Elderly

- Recommended dose: 7.5 mg/day
- If 15 mg/day is given initially, try to reduce the dose to 7.5 mg/day after the first 1–2 nights



Children and Adolescents

- Safety and efficacy have not been established
- Long-term effects of quazepam in children/adolescents are unknown

- Should generally receive lower doses and be more closely monitored

Pregnancy

- Contraindicated for use in pregnancy
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects on infant have been observed and include feeding difficulties, sedation, and weight loss

The Art of Psychopharmacology

Potential Advantages

- Transient insomnia

Potential Disadvantages

- Chronic nightly insomnia

Primary Target Symptoms

- Time to sleep onset
- Total night sleep
- Nighttime awakening

Pearls

* Because quazepam tends to accumulate over time, perhaps not the best hypnotic for chronic nightly use

- If tolerance develops, it may result in increased anxiety during the day and/or increased wakefulness during the latter part of the night
- Quazepam has a longer half-life than some other sedative hypnotics, so it may be less likely to cause rebound insomnia on discontinuation

* Long-term accumulation of quazepam and its active metabolites may cause insidious onset of confusion or falls, especially in the elderly

- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

Ankier SI , Goa KL . Quazepam. A preliminary review of its pharmacodynamic and pharmacokinetic properties, and therapeutic efficacy in insomnia . *Drugs* 1988 ;**35** :42 –62 .

.....

Hilbert JM , Battista D . Quazepam and flurazepam: differential pharmacokinetic and pharmacodynamic characteristics . *J Clin Psychiatry* 1991 ;**52** (Suppl):S21–6 .

.....

Kales A . Quazepam: hypnotic efficacy and side effects . *Pharmacotherapy* 1990 ;**10** :1 –10 .

.....

Kirkwood CK . Management of insomnia . *J Am Pharm Assoc (Wash)* 1999 ;**39** :688–96 .

.....

Roth T , Roehrs TA . A review of the safety profiles of benzodiazepine hypnotics . *J Clin Psychiatry* 1991 ;**52** (Suppl):S38 –41 .

Quetiapine

Therapeutics

Brands

- Seroquel
- Seroquel XR
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine, serotonin multimodal (DS-MM)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Acute schizophrenia in adults (quetiapine, quetiapine XR) and ages 13–17 (quetiapine)**
- **Schizophrenia maintenance (quetiapine XR)**

- **Acute mania in adults (quetiapine and quetiapine XR, monotherapy and adjunct to lithium or valproate) and ages 10–17 (quetiapine, monotherapy and adjunct to lithium or valproate)**
- **Bipolar maintenance (quetiapine, quetiapine XR)**
- **Bipolar depression (quetiapine, quetiapine XR)**
- **Depression (quetiapine XR, adjunct)**
- Other psychotic disorders
- Mixed mania
- Behavioral disturbances in dementias
- Behavioral disturbances in Parkinson's disease and Lewy body dementia
- Psychosis associated with levodopa treatment in Parkinson's disease
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Severe treatment-resistant anxiety
- Posttraumatic stress disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms

- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- * Interactions at a myriad of other neurotransmitter receptors may contribute to quetiapine's efficacy in treatment-resistant depression or bipolar depression, especially 5HT1A partial agonist action, norepinephrine reuptake blockade and 5HT2C antagonist and 5HT7 antagonist properties
- * Specifically, actions at 5HT1A receptors may contribute to efficacy for cognitive and affective symptoms in some patients, especially at moderate to high doses

How Long Until It Works

- Psychotic and manic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition and affective stabilization
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works –for Schizophrenia and Bipolar Disorder

- Most often reduces positive symptoms in schizophrenia but does not eliminate them